April Li

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Education

University of Waterloo **Expected Apr. 2029**
Bachelor of Computer Science, Co-op *GPA: 3.7/4.0*

Experience

Software Engineer May 2026 – Present
Ontario Power Generation *TypeScript, React, C#, ASP.NET Core, Azure, Docker, Kubernetes*

- Shipped ConstHub for **500+** engineers, digitizing **100+** operational workflows with electronic forms and approvals.
- Reduced manual form processing by **5x** with a schema-driven form engine using versioned JSON/HTML templates.
- Migrated **10K+** legacy forms and aggregated **250K+** records into interactive safety analytics covering criticality trends, deficiencies, and compliance metrics.
- Modernized Project Workbench from Angular to React/TypeScript and optimized .NET data access for **12K+** work orders, improving response times by **30%**.

Software Developer Jan 2026 – Mar 2026
Accelerate Alberta | GitHub | Live *React, Next.js, TypeScript, Resend, Vercel*

- Built and launched a production lead-capture and onboarding website, increasing online submissions by ∼**35%**.
- Achieved **95+** Lighthouse score and cut page load times from ∼**3s** to ∼**1.2s** through rendering and asset optimizations.

Software Developer Oct 2025 – Dec 2025
UW Electrium Mobility | *GitHub* *Next.js, TypeScript, Supabase, PostgreSQL, NextAuth*

- Streamlined project tracking for **50+** members across **7** subteams, eliminating ∼**40%** of manual coordination.
- Cut task-update latency by **80%** with real-time synchronization and optimistic UI updates across concurrent users.

Projects

CityWatch - HackCanada 2026 | *GitHub*
Python, TensorFlow, Keras, scikit-learn, OpenCV, FastAPI, PostgreSQL, GeoPandas, PostGIS

- Achieved **90.2%** accuracy across **10** infrastructure defect classes by preprocessing images with OpenCV and fine-tuning an ImageNet-pretrained MobileNetV2.
- Generated **30K+** synthetic infrastructure events and engineered **100+** features to train Random Forests forecasting 30-, 90-, and 365-day failure risk.
- Developed a FastAPI inference pipeline combining image, GPS, weather, and incident data into real-time infrastructure risk heatmaps across **100+** geographic regions.

NutriLens - Technova 2025 | *GitHub*
React Native, TypeScript, Expo, Node.js, MySQL, Gemini API

- Led **4** developers to build an AI food-safety platform analyzing barcode data across **3** nutrition sources in real-time.
- Achieved **95%** dietary-compatibility accuracy across **500+** product-diet pairs with Gemini to analyze ingredient lists, allergen data, and nutritional constraints.
- Implemented **sub-200 ms** product search across **250K+** foods with autocomplete, fuzzy matching, and dietary filters.

MailGest | *GitHub* | *Live*
React, Node.js, Express, MongoDB, Redis, GCP, Gemini API, OAuth

- Engineered an AI inbox pipeline processing **100+** emails per sync with priority classification and Gemini summaries.
- Reduced LLM calls by ∼**70%** and repeat-load latency by **2.5x** with incremental email detection and caching.
- Scaled LLM traffic with Redis rate limiting and async prefetching, improving responsiveness under request spikes.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C/C++, C#, SQL
Frameworks: React, React Native, Next.js, Node.js, Express, FastAPI, ASP.NET Core, Angular
Databases: PostgreSQL, MongoDB, MySQL, Supabase, Redis
Cloud & DevOps: Azure, AWS, GCP, Docker, Kubernetes, GitHub Actions
Machine Learning: TensorFlow, Keras, scikit-learn, OpenCV, NumPy, pandas
Certifications: AWS Certified AI Practitioner (Issued Sept 2025)